Lab 1

Sustainability & Green Tech

SRN: PES2UG25CS825

PROBLEM STATEMENT : Food Bank Surplus Redistribution Platfor

A platform that helps **restaurants and grocery stores donate surplus food** to **verified shelters** before it expires. Shelters can view, reserve, and schedule pickup of the available food.

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall allow donors to post surplus food packages with perishable expiry countdown timers and dietary tags.	High	Pass: Donor successfully posts a food package with expiry time and dietary tags. Fail: Required details are missing and the package cannot be posted.	Core functionality for listing surplus edible food.	Clear and testable.
FR-002	Functional	The system shall allow verified shelters to view available surplus food batches with quantity, expiry time, and dietary information.	High	Pass: Shelter can view available batches and their details. Fail: An expired batch is displayed as available.	Helps shelters identify suitable food before expiry.	Specify that only verified shelters can access claims.
FR-003	Functional	The system shall allow verified shelters to reserve available surplus food batches and select a pickup window.	High	Pass: Available batch is reserved and the pickup window is recorded. Fail: An already reserved or expired batch can be reserved.	Enables timely redistribution of surplus food.	Reservation should check availability.
FR-004	Functional	The system shall generate a unique pickup verification code for each confirmed food reservation.	Medium	Pass: A unique verification code is generated after successful reservation. Fail: A confirmed reservation has no verification code.	Verifies collection of the correct food batch.	Code should be linked to the reservation.
FR-005	Functional	The system shall automatically deactivate food batches when their expiry time is reached.	High	Pass: Expired batches are removed from the active claim board and cannot be reserved. Fail: An expired batch remains available.	Prevents expired food from being claimed.	Expiry handling must be automatic.
NFR-001	Nonfunctional	The platform shall dispatch instant push notifications to shelters within a 5 km radius when high-quantity perishable food is listed.	High	Pass: Eligible shelters within 5 km receive the notification within the defined notification target during testing. Fail: Eligible shelters do not receive it as required.	Helps perishable food reach shelters quickly.	Performance and notification security should be tested.
NFR-002	Nonfunctional	The platform shall restrict donor and shelter account information to authenticated and authorized users.	High	Pass: Only authenticated and authorized users can access protected information. Fail: An unauthorized user can access protected information.	Protects user information from unauthorized access.	Authentication and authorization should be verified.

2. Actors and Use Cases

Actors: Donor Partner, Shelter Coordinator, Platform Administrator

Primary Use Cases:

UC-01: Post Surplus Food;

UC-02: View Available Food;

UC-03: Reserve Food Batch;

UC-04: Schedule Pickup;

UC-05: Verify Pickup;

UC-06: Send Notifications;

UC-07: Check Food Availability;

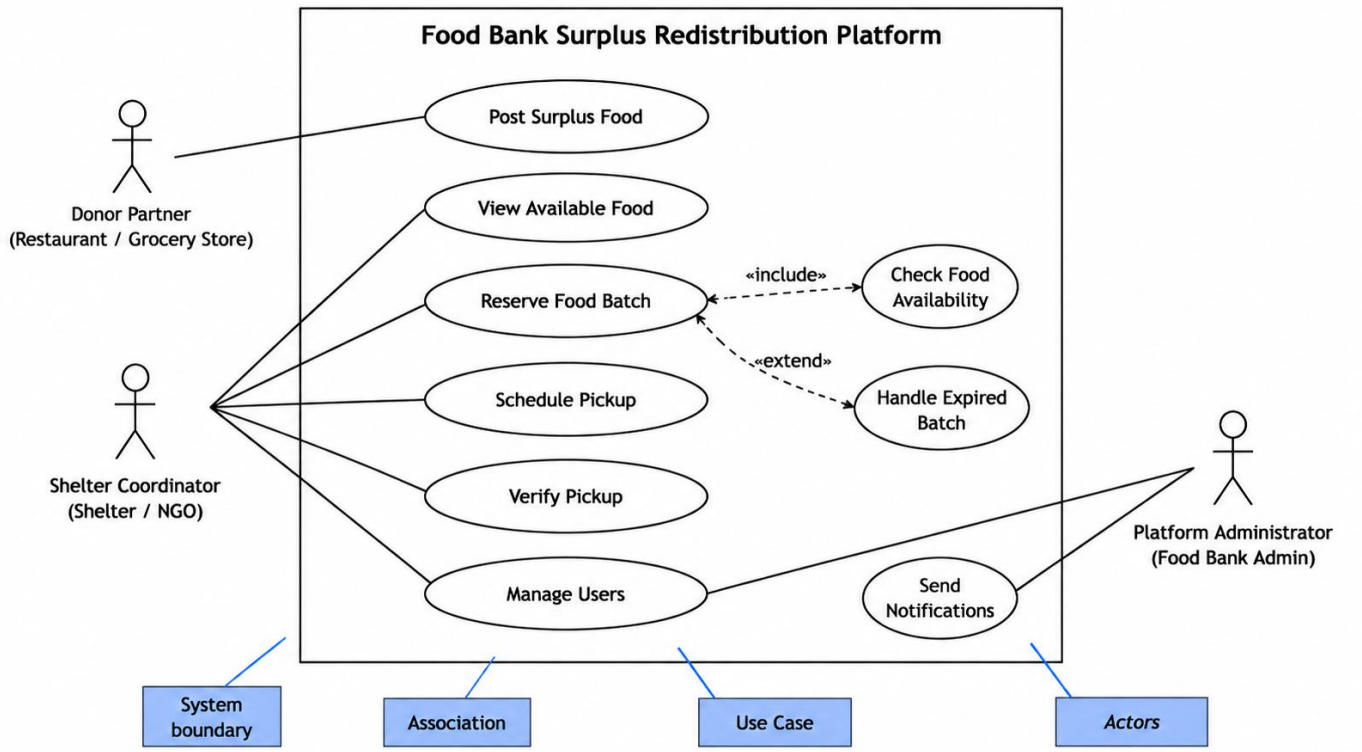
UC-08: Handle Expired Batch

Required relationships for the UML diagram:

• Reserve Food Batch «include» Check Food Availability

• Reserve Food Batch «extend» Handle Expired Batc

UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: UC-03 – Reserve Food Batch

Preconditions

- The Shelter Coordinator is a verified user.
- A surplus food batch is available.
- The food batch has not expired.

Postconditions

- The selected food batch is marked as reserved.
- The pickup window is recorded.
- A unique pickup verification code is generated.

Main Success Scenario

- Shelter Coordinator opens the available food list.
- System displays available surplus food batches.
- Shelter Coordinator selects a suitable food batch.
- System displays the batch quantity, expiry time, and dietary information.
- Shelter Coordinator selects a pickup window.
- System checks that the batch is still available.
- System reserves the food batch.
- System generates a pickup verification code.
- System displays the reservation confirmation.

Alternate Flow – Food Batch Expired

- Shelter Coordinator selects a food batch.
- System detects that the batch has expired.
- System prevents the reservation.
- System displays an “Food batch expired” message.
- Shelter Coordinator returns to the available food list.